LABORATORY REPORT · TSM2370 MATERIALTEKNIKK

Lab 1: Axial Tensile Testing and Strain Measurement Accuracy

Experiment title	Lab 1: Axial Tensile Testing and Strain Measurement Accuracy
Course / subject	TSM2370 Materialteknikk
Experiment no.	1
Date performed	*[DD/MM/YYYY]*
Date submitted	*[DD/MM/YYYY]*
Author / student name	*[First Last Name]*
Student ID	*[Student ID]*
Lab partner(s)	*[Lab partner names]*
Instructor / TA	Rasmus Ullestad Relling

1. Abstract

[A concise summary (150–250 words) of the whole report: the purpose, the methods, the main results with key numbers, and the main conclusion. Write the abstract last, when all other sections are finished.]

2. Introduction

2.1 Background and theory

Engineering stress (σ) is the applied axial force (F) divided by the original cross-sectional area (A_0). Engineering strain (ε) is the change in length (ΔL) divided by the original gauge length (L_0). The load cell of the universal testing machine (UTM) measures force accurately, but strain calculated from crosshead travel includes errors from the grips and the machine frame. An extensometer measures the elongation directly over the gauge length. The material goes from linear elastic deformation into plastic deformation, with strain hardening, necking and finally fracture.

Relevant formulas:

Engineering stress: $\sigma = F / A_0$

Engineering strain: $\varepsilon = \Delta L / L_0$

Young's modulus: $E = \sigma / \varepsilon$ ***(linear elastic region)***

2.2 Objectives

Determine how an unknown aluminium alloy behaves under a controlled, axial pulling load until fracture. The experiment also demonstrates why an extensometer is needed, instead of the machine's crosshead travel, to calculate strain accurately.

2.3 Hypothesis

[State your hypothesis or expected outcome, based on the theory.]

3. Materials and equipment

Device / material	Amount / specification	Manufacturer	Notes
Universal testing machine	10 kN (max. 5 kN grips)	Lloyd (LR 10K)	Axial pulling
Camera extensometer	1 unit	Basler / Logitech	Optical strain measurement
Aluminium sheet dog-bones	1 mm thick, 12.5 mm wide	ISO standard	Unknown alloy test coupons
Digital calipers	1 unit	–	For L_0, width and thickness

4. Safety considerations

4.1 Hazards

Pinch points in the UTM grips; sharp edges on fractured aluminium specimens.

4.2 Personal protective equipment (PPE)

Safety glasses, closed-toe shoes, standard lab clothing.

4.3 Waste disposal

Put the fractured aluminium pieces in the designated metal recycling bin.

5. Methods / procedure

1. Take safety precautions and make sure the UTM safety shields are in place.
2. Use digital calipers to measure the initial gauge length (L_0) and the width and thickness of the stamped aluminium part.
3. Enter the measured L_0 in the extensometer software and clamp the specimen securely in the UTM grips.
4. Start the tensile pull at the specified crosshead speed.
5. Run the test until the specimen fractures completely.
6. Export the raw force and elongation data from the LR 10K software.
7. Clean the workstation and dispose of the fractured aluminium pieces safely.

6. Results

6.1 Observations

[Describe any qualitative observations made during the experiment.]

6.2 Data table

Parameter / measurement	Value	Unit
Yield strength (machine travel)		MPa
Yield strength (extensometer)		MPa
Ultimate tensile strength (UTS)		MPa
Young's modulus (machine travel)		GPa
Young's modulus (extensometer)		GPa

6.3 Calculations

[Show one full sample calculation for each type of computation: the equation, the substituted values, and the result with correct units.]

6.4 Figures / graphs

[Insert figures or graphs here. Label the axes and give the units.]

7. Discussion

7.1 Interpretation of results

Discuss the difference in the linear elastic slope (E) between the two curves. Explain why the extensometer is required to get true material properties, and use the measured properties to determine the unknown aluminium alloy. *[Compare your values with literature values for common alloys, e.g. 6061-T6 and 5052-H32.]*

7.2 Comparison with theoretical values

[Discuss how your values compare with theoretical or literature values.]

7.3 Sources of error

Systematic errors: *[Identify calibration or method errors]*

Random errors: *[Identify environmental or measurement scatter errors]*

8. Conclusion

[Summarise the most important findings and answer the question posed in the objectives.]

9. References

- [1] Häll, K. (2026). Materials Science & Engineering: Practical Characterization. University of South-Eastern Norway.
- [2] USN Department of Science and Industry Systems (2026). TSM2370 Materialteknikk 2026 course material. University of South-Eastern Norway.
- [3] USN Kongsberg Mechanical Engineering Lab (2026). Equipment guides (Lloyd LR 10K, Matsuzawa MXT50). github.com/ravo10/USN-kongsberg-lab-mech

10. Appendices

Appendix A – Raw data

[Insert raw data logs or printouts here.]

Appendix B – Additional figures

[Insert additional photos or supplementary graphs here.]

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Lab 2: Thermal Processing and Quench Severity

Experiment title	Lab 2: Thermal Processing and Quench Severity
Course / subject	TSM2370 Materialteknikk
Experiment no.	2
Date performed	*[DD/MM/YYYY]*
Date submitted	*[DD/MM/YYYY]*
Author / student name	*[First Last Name]*
Student ID	*[Student ID]*
Lab partner(s)	*[Lab partner names]*
Instructor / TA	Rasmus Ullestad Relling

1. Abstract

[A concise summary (150–250 words) of the whole report: the purpose, the methods, the main results with key numbers, and the main conclusion. Write the abstract last, when all other sections are finished.]

2. Introduction

2.1 Background and theory

Hardness correlates strongly with a material's yield strength. The experiment uses two separate thermal units:

- **Primary austenitizing furnace (810 °C):** Samples B, C and D are all first placed in the primary furnace at 810 °C, inside the hardening range given in the datasheets for 1.2210 (780–840 °C), and held for 15–30 minutes. The matrix transforms to austenite (γ, FCC) and most of the carbon dissolves in the iron lattice. Because this steel has about 1.15 % carbon, a small amount of carbide stays undissolved. That is normal for a high-carbon tool steel and gives wear resistance; heating much higher would cause grain growth and a higher risk of cracking.
- **Quenching (cold water vs. warm water or oil):** Samples B and C are cooled rapidly, directly from 810 °C. Rapid quenching prevents carbon diffusion and forces the FCC austenite into a body-centred tetragonal (BCT) lattice: martensite, the hardest and most brittle microstructure. A 20 °C cold-water quench (Sample B) is more severe than 70 °C warm water or oil (Sample C). 1.2210 is a shallow-hardening steel, so in the slower quench of Sample C part of the austenite may transform to softer pearlite or bainite before martensite can form. Expect Sample C to be clearly softer than Sample B.
- **Isothermal annealing and air cooling (600 °C oven → ~20 °C air):** Sample D is NOT quenched in liquid. After austenitizing at 810 °C it is moved directly to a secondary oven at 600 °C and held for 15–30 minutes. Because 600 °C is well above the martensite start temperature (Ms), carbon diffuses quickly and the austenite transforms fully into soft, ductile pearlite (ferrite + cementite). When the transformation is complete, the sample is taken out and air-cooled on a heat-resistant surface to room temperature (~20 °C) for handling and hardness testing.

Relevant formulas:

Vickers hardness: $HV = 1.8544 \cdot F / d^2$

Mean diagonal: $d = (d_1 + d_2) / 2$

Units: ***F in kgf, d in mm. This lab uses HV1 (F = 1 kgf).***

2.2 Objectives

Measure the local resistance to plastic deformation (Vickers hardness) of different microstructures in Sølvstål 2210 (1.2210 / 115CrV3, a high-carbon tool steel). The experiment shows how the quench medium (cold water vs. warm water or oil) and the thermal processing route decide the final mechanical properties.

2.3 Hypothesis

[State your hypothesis or expected outcome, based on the theory.]

3. Materials and equipment

Device / material	Amount / specification	Manufacturer / grade	Notes
High-carbon steel blocks	Prepared pieces (Sølvstål 2210)	1.2210 / 115CrV3	4 prepared samples per group
Lab furnaces and ovens	Primary furnace (810 °C) and secondary oven (600 °C)	–	Austenitizing (810 °C) and isothermal anneal (600 °C)
Quench media stations	20 °C water, 70 °C water, oil	–	Different quench severity
Vickers hardness tester	1 unit	Matsuzawa (MXT50)	HV1 (1 kgf), 15 s dwell
Polishing equipment	As needed	–	Prepare surfaces for indentation

4. Safety considerations

4.1 Hazards

Extreme heat (810 °C furnace, 600 °C oven), hot steel blocks, burns, quenching vapours and splashes. High-carbon steel can crack during a severe water quench. Hot quenching oil can catch fire if a hot sample is lowered in too slowly or the oil is too shallow: keep the lid within reach and never use water on an oil fire.

4.2 Personal protective equipment (PPE)

Heat-resistant gloves, tongs, face shield, lab coat, closed-toe shoes.

4.3 Waste disposal

Dispose of polishing waste properly. Handle quenching fluids according to the lab's environmental guidelines.

5. Methods / procedure

1. Take safety precautions and put on heat-resistant protective equipment (gloves, tongs, face shield). Keep the lid for the oil bath within reach.
2. **Sample A (baseline):** Measure the baseline Vickers hardness of the prepared, untreated steel block.
3. **Sample B (cold-water quench):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Quench rapidly in 20 °C cold water, moving the sample around in the bath with tongs. Polish and measure the Vickers hardness.
4. **Sample C (reduced-severity quench):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Quench in 70 °C warm water OR quenching oil. Polish and measure the Vickers hardness.
5. **Sample D (600 °C isothermal anneal and air cooling):** Place in the 810 °C primary furnace and hold for 15–30 minutes. Move it directly to the secondary oven at 600 °C and hold for 15–30 minutes for complete pearlite transformation. Take it out and let it air-cool on a heat-resistant surface to room temperature (~20 °C). Polish and measure the Vickers hardness.
6. Record all hardness indentations and clear the testing equipment.

6. Results

6.1 Observations

[Describe any qualitative observations made during the experiment.]

6.2 Data table

Parameter / measurement	Value	Expected (approx.)	Unit
Sample A hardness (baseline, untreated)		220–240	HV1
Sample B hardness (810 °C → 20 °C water quench)		750–850	HV1
Sample C hardness (810 °C → 70 °C water / oil quench)		450–750	HV1
Sample D hardness (810 °C → 600 °C anneal → 20 °C air)		250–350	HV1

6.3 Calculations

[Show one full sample calculation for each type of computation: the equation, the substituted values, and the result with correct units.]

6.4 Figures / graphs

[Insert figures or graphs here. Label the axes and give the units.]

7. Discussion

7.1 Interpretation of results

Compare Sample B (cold water) and Sample C (warm water or oil) to explain quench severity and cooling rate in high-carbon tool steel, and whether Sample C hardened fully. Compare Sample B (martensite) and

Sample D (810 °C → 600 °C for 15–30 min, then air-cooled) to explain how the cooling route controls the phase transformation (BCT martensite vs. pearlite) and the final hardness.

7.2 Comparison with theoretical values

Expected values (guide only): Sample A 220–240 HV1, Sample B 750–850 HV1, Sample C 450–750 HV1 (below B), Sample D 250–350 HV1 (above A). A is based on the delivery condition in the datasheets (max. about 220 HB, roughly 230 HV) and a measurement on an untreated sample in the lab (236 HV1). B is based on the datasheet hardness after hardening (64–66 HRC, roughly 800–865 HV), a little lower here because the samples are small, not tempered and measured near the surface. C depends strongly on the quench medium and on how fast the sample gets into the bath. Conversions between HB, HRC and HV follow ISO 18265 and are approximate. *[Discuss how your values compare with these, and explain any differences.]*

7.3 Sources of error

Systematic errors: *[Identify calibration or method errors]*

Random errors: *[Identify environmental or measurement scatter errors]*

8. Conclusion

[Summarise the most important findings and answer the question posed in the objectives.]

9. References

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